

# **ENVIRONMENTAL EDUCATION**

(Text-book based on the New Syllabus  
introduced in 2021 by the  
Tamilnadu Teacher Education University, Chennai)

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## **PREFACE**

This text book titled 'Environmental Education' has been written to cover the new syllabus of B.Ed. Course of the Tamilnadu Teacher Education University introduced in 2021. The five units in the new syllabus are presented under the following headings:

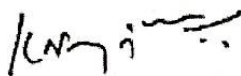
(1) Environmental Education (2) Natural Resources, Problems and Solutions (3) Environmental Pollution, Hazards and Disaster Management (4) Environmental Problems, Policies and Protection of Environment (5) Environmental Education in the School Curriculum.

To facilitate easy understanding of the subject by the teacher-trainees, the contents have been presented under different headings and sub-headings. As of our previous works, this book is also written in a lucid style with vivid explanations and apt examples. To help students prepare for their examinations, review questions and the expected answers for the same have been provided at the end of each unit. A set of Model Question Papers based on the University examination pattern with the expected answers for the same cited in the book, have been provided at the end of this book to help students who are to take their B.Ed. Degree Examination under the new syllabus for the first time.

I am thankful to, **Mr.C.R. Manivasagan, M.C. n., M.Ed.**, former Office Superintendent of Meston College of Education, Chennai – 14, for his pains-taking efforts in proof reading this book and **Maha Agencies** for neatly printing this book.

Constructive suggestions are quite welcome to refine this book further. I earnestly solicit the continued support of teacher-educators and student-teachers, as they have been offering for my previous works. It will be highly satisfying to me if this book proves useful in the hands of student community.

Chennai – 87  
09-03-2022



(Dr.K.NAGARAJAN)

1. Symbols (A), (B), (C) used in Review Questions refer to Essay, Short Answer and Very Short Answer type questions respectively.
2. Essay type questions are to be answered in about 5 pages.
3. Short Answer type questions are to be answered in about 2½ page.

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## Unit - I

### ENVIRONMENTAL EDUCATION

#### 1:00 Introduction

The magnificent mountains, the lushy green forests, rapturous rivers, the deep oceans and seas, the blue lagoons and lakes, marvelous reservoirs of the underground wealth ..... water and minerals, the bewildering varieties of animals, the beautiful birds, several insects, pests, tiny and invisible micro-organisms ... all these constitute our planet earth. Interactions between matter and energy among various living species decide the composition of this planet and ecological balance at a point of time.

Due to man's interaction with nature on a large scale, the balance of nature has been upset and environmental degradation has occurred in most parts of the world, because of environmental pollution, improper and unscientific exploitation of natural resources etc. It has posed a great danger threatening the very existence of man, plant and animal life on the earth. So, there is a need to increase awareness and understanding of those environments and man's impact on them and to find out the effective ways to manage them. To achieve

the above goal, environmental education is the need of the day.

In this unit, Meaning of the term environment, its definition, Types and components of environment, Environmental awareness, Environmental attitude, Ecological intelligence, Ecological sensitivity, Meaning and definition of environmental education, Characteristics of environmental education, Focal aspects of environmental education, Goals and objectives of environmental education, Need and importance of environmental education and Scope of environmental education are taken up for discussion.

### 1:01 Meaning of Environment

The dictionary meaning of the word 'Environment' is a surrounding; external conditions influencing development or growth of people, animals or plants; living or working conditions etc. This involves three questions: (i) What is surrounded? (ii) By what surrounded? and iii) Where surrounded? The answer to the first question is living beings in general and man in particular. If man is taken to be surrounded, physical attributes of the surrounding, become the answer for the second question, which

become then 'environment'. The space or habitat is the answer for the third question. Thus **'Environment refers to the sum total of conditions which surround man at a given point in space and time'.**

Primarily, the concern of all educationists is with the environment of man. But man cannot exist or be understood in isolation from the other forms of life and so the environment of all biological population should be the concern of teachers.

Generally speaking 'environment' is equated with nature wherein physical components of the planet earth, viz. land, water, air etc. that support and affect life in biosphere. **A. Goudie** (1984) in his book "**The Nature of the Environment**" has taken environment as the representative of physical components of the earth wherein man is an important factor affecting the environment".

Environment is defined more comprehensively by others as holistic view of the world as it functions at any point of time with a multitude of spatial elements and socio-economic systems distinguished by quality and attributes of space and mode of behaviour of physical and biological forms. But Goudie pleads in his book, "the definition and in turn the scope, could be governed by our concern and

priorities. Our immediate concern is the quality of space we live in, the air we breathe, the food we eat, the water we drink and the resources we draw from the environment to support our economy. As such only 'air-land-water-plant' are to be included in the concept of environment, thus excluding man and human society from the ambit of environment.

### **1:02 Definition of 'Environment'**

The child when it comes to this world, finds itself surrounded by innumerable objects and circumstances which influence it. All these except the child form the environment. Given below are some of the definitions for 'environment'

'Environment is anything immediately surrounding an object and exerting a direct influence on it

**- P. Gisbert**

"A person's environment consists of the sum total of the stimulation which he receives from his conception until his death"

**- Boring**

"Environment is the field of effective stimulation and interaction for any unit of living matter"

**- T.D. Elliot**

"Environment is everything that affects the individual except his genes"

**- Anastasi**

"Environment is an external force which influence us"

**- E.J. Ross**

"The term environment is used to describe, in the aggregate, all the external forces, influences and conditions which affect the life, nature, behaviour and the growth, development and maturity of living organisms"

**- Douglass and Holland**

### **1:03 Types of Environment and its Components**

There are two kinds of environment - (i) Natural Environment and (ii) Man-made Environment.

#### **1:03:1 Natural Environment**

This consists of air, water, land, forests, mountains, radiations, living organisms such as animals, birds, reptiles and micro organisms like bacterias, viruses and fungus as its components. It has a homeostatic mechanism i.e. any change in one component of the ecosystem is counter balanced by changes in the other components of the environment. Thus different components of the environment are independent as well as interdependent, maintaining a reciprocal relationship among themselves.

The natural environment could be further classified as follows :

- i) Biotic or living component
- ii) Abiotic or Physical Component

### **1:03:1 :01 Components of Biotic Environment**

The biotic component of the environment could be classified as those organisms living on the land (Terrestrial) and those living in water (Aquatic). Terrestrial organisms could be further sub-divided into three kinds (i) Flora (Vegetations like Plants) (ii) Fauna (Animals) and (iii) others.

#### **Flora**

It consists of different species of vegetations like plants, grass, bushes, forests etc. which do not have the ability to move.

#### **Fauna**

It consists of mammals and other species like birds, reptiles, fishes, sharks, whales etc. These organisms have the ability to move by themselves.

#### **Others**

This consists of micro-organisms like bacterias, viruses, fungus etc.

### **1:03:1:02 Components of Abiotic Environment**

Abiotic or Physical environment has three components. They are (i) Lithosphere (ii) Hydrosphere and (iii) Atmosphere.

#### **i) Lithosphere**

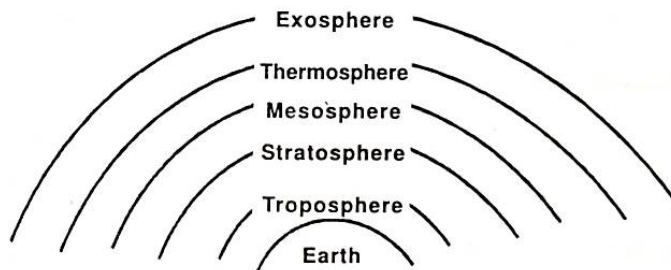
This refers to the crust of the earth constituted by rigid rocks and soil. Soil is a complex mixture of organic and inorganic matter as well as air and water.

#### **ii) Hydrosphere**

It consists of Oceans, seas, rivers, streams, lakes, polar ice-caps, glaciers, ground water etc.

#### **iii) Atmosphere**

Atmosphere is a gaseous envelope extending to a height of 20,000 K.M. from the earth surface. It consists of Nitrogen, Oxygen, Carbondioxide, Hydrogen, Argon etc. There are 5 layers in the atmosphere as shown below :



(Five layers of Atmosphere (Not according to Scale))

## **Troposphere**

This is the bottom layer of the atmosphere which extends to a height of 10-15 K.M. from the surface of the earth. In this layer, as the height increases, temperature decreases and reaches  $-60^{\circ}\text{C}$  level.

## **Stratosphere**

This layer just above 'troposphere' extends from 15 to 50 K.M. of height from the surface of the earth. In this layer, as the height increases, temperature also increases and reaches the maximum temperature of  $+90^{\circ}\text{C}$ .

## **Mesosphere**

This layer lying above 'Stratosphere' extends from 50 to 85 K.M. of height from the surface of the earth. In this layer, as the height increases, temperature decreases and reaches  $-80^{\circ}\text{C}$  level.

## **Thermosphere**

This layer of atmosphere is lying at 85 to 400 K.M. of height from the surface of the earth and the temperature in this layer is above  $1000^{\circ}\text{C}$ .

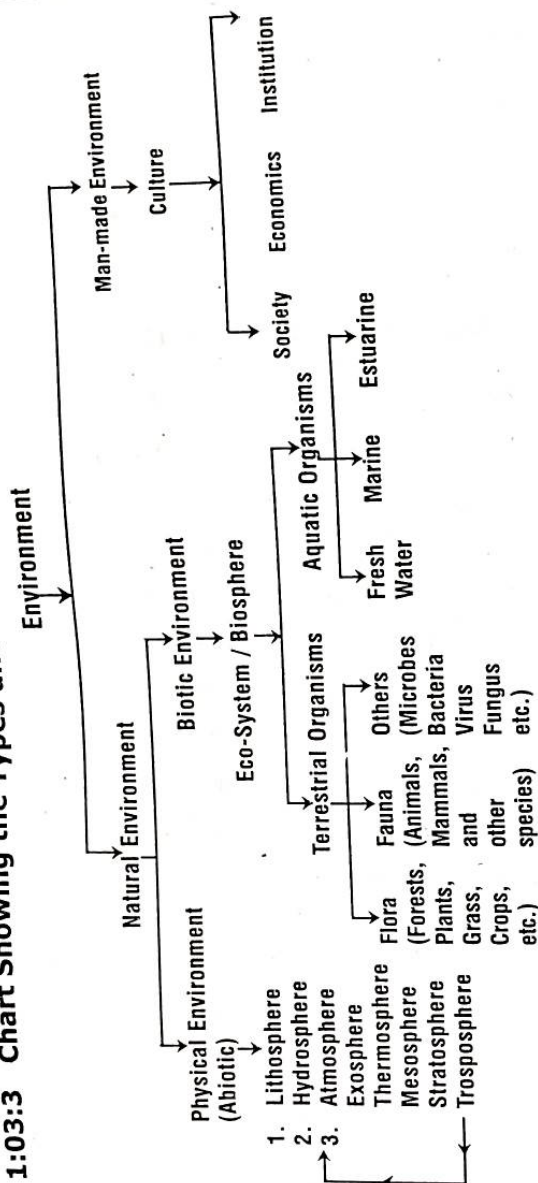
## **Exosphere**

This is the topmost layer of atmosphere lying at 400 to 20,000 K.M. of height from the surface of the earth where temperature ranges from  $1000^{\circ}\text{C}$  to  $10,000^{\circ}\text{C}$ .

## **1:03:2 Man-made Environment**

This refers to the artificial environment created by man. Man has superior modern technologies with the help of which he could create an environment as desired by him. Man is the most potent agent of environmental intervention. He is able to expand transport facilities, build dams and extend the irrigational areas, increase housing facilities, produce and use nuclear, hydroelectric and thermal power, etc. with the help of advancement in science and technologies. This involves Social, Cultural and Institutional components.

**1:03:3 Chart Showing the Types and Components of Environment**



## 1:04 Environmental Awareness

The terms Environmental education and environmental awareness are used interchangeably for the same meaning but there is significant difference in these two terms. The study of physical and biosciences, geography, agriculture etc., provides the environmental awareness. But the awareness does not help in developing skills and attitudes for improving environment. Therefore it is essential to understand the concept of environment and environmental awareness and differentiate it with environmental education.

**Environmental awareness** may be defined to help the social groups and individuals to gain a variety of experiences in and acquire a basic understanding of environment and its associated problems. Educators and Environmental specialists have repeatedly pointed out that any solution to the environmental crisis will require environmental awareness and understanding to be deeply rooted in the educational system at all levels.

## 1:05 Environmental Attitude

Attitudes are learned dispositions towards aspects of our environment. They may be positively

or negatively directed towards certain issues, people or institution. Environmental attitude refers to one's attitude towards preservation of the quality environment and utilization of environmental resources.

Environmental attitudes are important because they often, but not always, determine behaviour that either increases or decreases environmental quality. Traditionally, attitudes have cognitive, affective, and conative elements, but environmental attitudes might be better described as having preservation and utilization dimensions. Pro-environmental attitudes rise and fall with current events and vary with age, gender, socioeconomic status, nation, urban-rural residence, religion, politics, values, personality, experience, education, and environmental knowledge. Environmental education aims to improve environmental attitudes but has mixed results. The mass media have been both helpful and harmful. Two prominent theories for explaining environmental attitude-behaviour relations are the theory of planned behaviour and value-beliefs-norm theory, which offer the benefit of parsimony and the shortcoming of incompleteness. Researchers have, for example, suggested additions to the theory of planned

behaviour, noting that pro-environmental behaviours vary in their effort to complete, which influences the attitude-behaviour relation, and that many barriers to behaviour-change exist.

### 1:06 Ecological Intelligence

**Ecological intelligence denotes one's knowledge, understanding and attitudes developed towards environment and maintaining its balance.**

Ecological intelligence help us to comprehend different natural systems in all their complexity as well as the interplay between natural and man-made worlds.

Ecologists tell us that natural systems operate on multiple scales. At the macro-level there are global biogeochemical cycles like that for the flow of carbon, where shifts in the ratios of elements can be measured not just over the years but over centuries and geologic ages. The echo systems of a forest balances the interplay of plants, animals and insect species', down to the bacteria in the soils each finding an ecological niche to exploit, their genes evolving together. At the micro-level, cycles run their course on a scale of millimeter or microns, in just seconds.

How we just perceive and understand all these makes a crucial difference and reason for their difference is what we refer 'Ecological intelligence'. "The huge tree which moves some people to tears of joy, is in the eyes of others only a green thing which stands obstructing the way", wrote the poet **William Blake**, two centuries ago, "some see nature all ridicule and deformity, and some scarcely see Nature at all. But to the eyes of the man of imagination, Nature is imagination itself. As a man, so he sees".

The store of knowledge about different ecological systems, with all their complexities, as well as the interplay between natural and man-made worlds and their resultant impact is huge that no single human brain can store it all. The findings of different people at different periods of times, about the different natural eco-systems, their complexities, the interplay among them and their impact, maintaining the balance between different natural systems etc., constitute the vast store of ecological knowledge. Each one of us needs the help of others to navigate this huge knowledge by which only each individual can develop his/her ecological intelligence.

Psychologists conventionally view intelligence as

residing within an individual. But the ecological abilities we need to survive today must be a '**Collective intelligence**'; one that we learn and master as a species and that resides in a distributed fashion among the far-flung net-works of people. How far one has acquired this collective intelligence denotes his ecological intelligence.

### 1:07 Ecological Sensitivity

**Environmental sensitivity** describes the ability of an individual to perceive and process information about their environment. It is a fundamental and basic trait found in many organisms and is crucial because it enables an individual to adapt to different environmental conditions. Levels of Environmental Sensitivity often vary considerably from individual to individual, with some being more and others less sensitive to the same conditions. Such differences have been observed across many species such as pumpkinseed fish, zebra finches, mice, non-human primates and humans, indicating that there is a biological basis to differences in sensitivity.

### 1:08 Meaning and Definition of Environmental Education

Environmental Education, which is relatively new,

is an interdisciplinary process that aims at equipping people with the knowledge, attitudes, skills and motivation that they need to help resolve environmental issues. Unlike the formal education, Environmental Education aims at effecting behavioural change. Knowledge acquired through Environmental Education should lead to change in people's values and attitudes. Some of the definitions have been provided to understand the concept.

1. "Environmental education is the process of recognizing values and clarifying concepts in order to develop skills and attitude necessary to understand and appreciate the interrelatedness among man, his culture and his biophysical surroundings. It also entails practice in decision making and self formulation of a code of behaviour about problems and issues concerning environmental quality" – **UNESCO (1970) Working Committee**
2. "Environmental education is a way of implementing the goals of environmental protection. It is not a separate branch of science or field of study. It should be carried out according to the principles of life long integral education". – **UNESCO (1976) Seminar at Jammi**

3. "Environmental education appears to be a process that equips human beings with awareness, knowledge, skills, attitudes and commitment to improve environment". – **Mishra (1993)**
4. "Environmental education refers to the awareness of physical and cultural environment and perceive its relevance for real life situation. The problems and issues are to be identified. The imbalances of environment are to be improved in view of sustainable development". – **R.A. Sharma (1996)**
5. "Environmental education is problem-centred, interdisciplinary, value-oriented, community-oriented, and concerns with man's survival as species, based on student-initiated activities and involvements present and future oriented" – **Cook and Hearn, 1971**

### **1:09 Characteristics of Environmental Education**

From the above stated definitions the nature (characteristics) of Environmental Education could be summarised as under:

- i) Environmental education is a process of providing learning experiences to obtain knowledge, understanding, skills and

awareness with desirable attitudinal changes about man's relationship with his natural and man made surroundings which includes the relation of population, pollution, resources allocation, transportation technology and urban and rural planning to the total human environment.

- ii) Environmental education must utilize diverse learning environments and a broad array of educational approaches to teaching learning about and from the environment with due stress on practical activities and first hand experience.
- iii) Environmental education should help learners to discover the symptoms and real causes of environmental problems and thus to develop critical thinking and problem solving skills. In Environmental Education one has to look at various aspects of environmental problems objectively and take decisions accordingly. Very often, the decision may be in conflict with the present practices and beliefs.
- iv) Environmental education also develops, modifies and improves upon existing attitudes and values towards one's environment.
- v) Environmental education should be a continuous life-long process, beginning at the

pre-school stage level and continuing through all **formal and non-formal stages** and

- vi) Environmental education should be interdisciplinary discipline in making possible a holistic and balanced perspective. This is because environmental problems and issues can not be confined to the boundaries of physics, chemistry, botany, zoology or any discipline. On the other hand, Environmental Education is an amalgamation of many of the conventional subjects.

### **1:10 Focal Aspects of Environmental Education**

In line with the core themes of Environmental Education, the six focal aspects and its components are listed below:

#### **1. Natural resources, Problems and Solutions**

- i) Inexhaustible resources and Non-renewable resources
- ii) Natural resources such as land, water, air, minerals, forests, food and energy resources.
- iii) Problems related to Natural resources:
  - a) Land Resources: Landslides, Soil-erosion, Loss of fertility, Desertification.
  - b) Forest Resources : Deforestation.

- c) Water Resources: Over-exploitation of ground water, Pollution due to Industrial Effluents, Sewage and Agricultural discharges etc.
  - d) Mineral Resources: Over-exploitation of Mineral Resources
  - e) Food Resources : Decrease in food production due to Farm land conversion, Mono crop culture, over fishing etc.
  - f) Energy Resources : Depletion of Fuel resources due to increasing demand for energy, not tapping the Non-conventional resources fully.
- (iv) Role of Individual in conserving the natural resources:
- a. Conserving the bio-diversity
  - b. Energy saving
  - c. Reducing the usage of paper
  - d. Avoiding plastic items
  - e. Other ways of conserving natural resources

## 2. Environmental Pollution and Hazards

- (i) Environmental Pollution
  - a) Land pollution
  - b) Water pollution
  - c) Air pollution

- d) Noise pollution
  - e) Light pollution
- (ii) Various environmental hazards and disasters
- a) Earthquake
  - b) Tsunami
  - c) Landslide
  - d) Volcano explosion
  - e) Floods
  - f) Cyclone
  - g) Forest fire
  - h) Nuclear accidents
  - i) Industrial accidents

## 3. Important environmental problems

- (i) Global Warming
- (ii) Green House Effect
- (iii) Depletion of Ozone layer
- (iv) Acid Rain
- (v) Extinction of Flora and Fauna

## 4. Environmental Conservation Policies and Programmes

- (i) Efforts taken in India for the conservation of environment and the laws enacted in this regard
- (ii) Creation of Gene Banks
- (iii) Solar Power Projects
- (iv) Rain Water Harvesting
- (v) Development of Social Forests

## 5. Environmental Management and Protection

- (i) Soil Management
- (ii) Water Management
- (iii) Waste Management
- (iv) Disaster Management
- (v) Establishment of International organisations for the Protection and Conservation of Environment.
- vi) Environmental Movements in India and the functioning of NGO's in the Protection of Environment.

## 6. Incorporation of 'Environmental Education' in the School Curriculum to develop environmental awareness, attitude and skills for solving environmental problems.

- (i) Providing environmental education at various levels of education
- (ii) Approaches in providing environmental education
- (iii) Ways and Means of integrating environmental education in the curriculum
- (iv) Guidelines to include environmental education at school level

- (v) Problems faced in imparting environmental education

## 1:11 Goals of Environmental Education

**UNESCO has highlighted the following aims of environmental education:**

The aim of environmental education is clearly to show the economic, social, political and ecological interdependence of the modern world, in which decisions and actions by different countries can have international repercussions. Environmental education should, in this regard, help to develop a sense of responsibility and solidarity among countries and regions as the foundation for a new international order which will guarantee the conservation and improvement of the environment.

The main aim of environmental education at the grass root level is to succeed in making individuals and communities understand the complex nature of the natural and the built environments. Further, to acquire the knowledge, values, attitudes, and practical skills to participate in a responsible and effective way in anticipating and solving social problems, and in the management of the quality of the environment.

Therefore, necessary steps for providing environmental education are:

### Developing in students

- (a) Awareness;
- (b) Knowledge;
- (c) Attitude building for motivating to protect environment;
- (d) Evaluation of environmental measures; and
- (e) Skill and capacity building!

According to **D.H. Meadows'**, environmental educators on every continent develop materials and methods as varied as the different cultures and ecosystems on earth. He lists some key concepts which underlie all environmental education. These are food for thought, levels of being, complex systems, population growth and carrying capacity, ecologically sustainable development, socially sustainable development, knowledge, uncertainty and sacredness.

### 1:12 Objectives of Environmental Education

There are several governmental and non-governmental organizations to educate people and to create awareness of environment. The course content of Environmental Education (EE) has been designed for both, the formal system and non-formal system of education. A department of environment

has been set up in 1982 on Environmental information system for this purpose. There is a Centre for Environmental Education (CEE) at Ahmedabad. There are more than two hundred private organizations working for environmental education.

A number of objectives of environmental education have been formulated for both formal and non-formal education at all levels at the International Conference of UNESCO (1977) held at Tbilisi, These are as follows :

- (1) To develop an awareness of environment and sensitivity (feeling and attitudes) to the total environment and its allied problems.
- (2) To help acquiring knowledge and variety of experiences of the environment and associated problems.
- (3) To develop a basic understanding of structure, processes and problems of environment, interdependence of environmental components.
- (4) To help in acquiring skills for identifying and solving environment problems.
- (5) To develop attitudes, a set of values and feelings of concern for the environment and encouragement or motivation for active participation in protection and improvement of environment.

- (6) To provide an opportunity for an active participation or practice at all levels in working for the solution of environmental problems and
- (7) To develop an ability for evaluating environmental components of educational programmes in terms of ecological, economic, social, cultural, aesthetic and educational factors. The above objectives are related to all levels and for both formal and non-formal system of education. In the formal system of education, four different but interrelated components have been recognized which are noted as follows.

| <i>Objectives</i>                                                                                | <i>Level</i>                     |
|--------------------------------------------------------------------------------------------------|----------------------------------|
| 1. Awareness of environment (Knowledge)                                                          | Primary education                |
| 2. Relevance for real-life situations of environment ( <i>understanding</i> )                    | Secondary education              |
| 3. Conservation of natural resources of environment ( <i>skills</i> )                            | Higher Secondary Education       |
| 4. Sustainable development by solving problems of environment ( <i>Attitude and Evaluation</i> ) | College and University Education |

Thus objectives of environmental education involve the understanding and doing aspects of environment.

### 1:13 Need for Environmental Education

Some form of environmental pollution affects each and every nation though the issues differ. Environmental issues differ from developmental issues in their impact upon individual lives. Environmental stress may or may not touch people's lives, in ways which are conscious and about which they can protest, but they are always the sum total of living things, the biosphere. Acid rain, desertification, global warming, ozone layer depletion, pollution of air, water and soil, radioactive contamination of large areas, and species extinction are some of the most urgent environmental threats to be dealt with now and in the future. These problems are serious, interdependent, and characterized in most cases by a dimension in space and time.

For more than twenty years, educational institutions all over the world have been making intense efforts to meet the challenges arising from politics, science and public opinion towards integrating environmental concerns. A popular slogan of the international ecology movement is "**think globally and act locally**".

Today, environmental education is an important segment within the educational system. In some

countries, it also constitutes a political-pedagogical action programme to be developed and pursued by social groups, government, the scientific community and educational institutions. This programme includes all educational activities consciously confronting and attempting to overcome the environmental crisis. It is a programme with the following aspects:

- ◆ Goals, concepts and components of the specific educational measures.
- ◆ Relationship between environmental crises and environmental education.
- ◆ Subjects and actors – teachers and pupils and their qualifications.
- ◆ Target groups.
- ◆ Local environment.
- ◆ Educational institutions.
- ◆ Conditions and access and participation.

Environmental education has a single, clearly defined and multifaceted object: the environmental crisis. Environmental learning is learning about the factors, causes and solutions to environmental crises. Learning about the environment is immediate reaction to concrete problems in the management of natural resources.

Present civilization finds itself in a state of development in which environmentally responsible behaviour is no longer the norm and the daily routine, unlike the traditional societies. Such behaviour is no longer learned and internalized naturally but must be taught consciously.

### **1:14 Importance of Environmental Education**

Over recent decades, global problems relating to degradation of natural resources and pollution have increased dramatically. Natural resources are depleted by excessive use. Fresh water scarcity on a global scale, deforestation, degradation of coastal and marine areas, soil erosion and loss of biodiversity, are some of the problems that have become a cause for concern. Air and water pollution has reached levels that are already resulting in serious health problems, as well as a negative impact on the environment, and inevitably influencing prospects for long-term economic growth. In such a scenario, the importance and need for environmental education can hardly be stressed. In fact, sustainable development or economic development without affecting the natural resources should form the cornerstone of environmental education.

The concept of sustainable development holds the vision of a more equitable world. This can be

achieved by providing the disadvantaged with the means to advance themselves and their families. One of these means, of course, is environmental education.

As universal literacy is still a distant dream, orienting environmental education towards sustainable development alone is not enough. Attention also has to be given to those who are at present not served or are poorly served by schools. This is a large group, well over a billion people and a vital one for the future. Women and girls, the mothers of today and tomorrow, are in the majority. They are, or will be, the first and most influential teachers of their children.

Environmental education is both valuable and necessary. Starting from very young age, children should be taught about the environmental awareness and knowledge of the area in which they live, should also grow. Their education should be integrated with core disciplines.

Environmental education provides the foundation for all future education and learning. It can produce children who are not only happy but are also happy with others, find learning exciting and develop enquiring minds. Environmental education enables children to develop a storehouse of knowledge about

the world and seek knowledge that they can use and develop throughout their lives. Environmental education empowers adults by enabling them to participate in a sustainable future. Environmental education thus lays the foundation for a lifelong learning.

In effect, the importance of environmental education can be stated as follows:

- ◆ It is essential for the self-fulfillment and social development of the child and the adult.
- ◆ It is essential for understanding the different food chains and the nature's ecological balance.
- ◆ It plays an important role in understanding and appreciating how the environment is used for making a living and promoting material culture.
- ◆ It enables one to appreciate and enjoy nature and society.
- ◆ It inculcates a concern for the systematic change of environment for the distant and the immediate welfare of mankind.
- ◆ It makes one conscious of the problems of population explosion, depletion of natural resources, global warming etc.

### **1:15 Scope of Environmental Education**

The scope and content of Environmental

Education can be studied in terms of four components :

- ◆ Awareness
- ◆ Real Life Situations
- ◆ Conservation
- ◆ Sustainable Development

**Awareness :** In this segment, an effort is made to make the individual conscious about physical, biological and cultural aspects of environment. The environment is linked with the life support system comprising 6 components: 1) air, 2) land, 3) water, 4) flora, 5) fauna, and 6) sunlight.

**Real Life Situations :** These situations link environment to life. As conditions are location-specific, problems and priorities of each area are different.

**Conservation :** This encompasses utilization of natural resources, not only by the present but also by the future generations. It does not include the process of exploitation.

**Sustainable Development :** This process aims at the sensible utilization of resources for development. All resources are finite and there is also a limit to the growth of the living system. In effect, one should utilize the limited natural resources wisely and with intelligence.

Regarding the scope of Environmental Education, the International Conference on Environmental Education putforth the following guidelines.

- ◆ The environment should be considered in its totality, natural and built, technological and social.
- ◆ Environmental Education should be a continuous lifelong process, starting at the preschool level and continuing at all levels of education, whether formal or non-formal.
- ◆ Environmental Education should adopt an interdisciplinary approach drawing on the content of each and enabling a holistic and balanced perspective.
- ◆ Students should be enabled to get insights into environmental conditions in other countries by examining major environmental issues from local, national and international points of view.
- ◆ While historical perspectives should be taken into consideration, the focus should be on the current and potential environmental issues.
- ◆ Promote international cooperation in the prevention and solution of environment problems.

- ◆ Ensure that environmental aspects have a place in plans for development and growth.
- ◆ Facilitate learners to have a role in planning their learning experiences.
- ◆ Enable learners to discover the symptoms and real causes of environmental problems.
- ◆ Relate environmental sensitivity, knowledge, problem-solving skills and value clarification to every age, but with special emphasis on environmental sensitivity to the learner's own community in early years.
- ◆ Stress the complexity of environmental problems and the need to develop critical thinking and problem-solving skills.
- ◆ Utilize diverse learning environments with due stress on first-hand experiences and practical activities.

## Conclusion

In this unit, meaning of the term environment, its definition, types and components of environment, environmental awareness, environmental attitude, ecological intelligence, ecological sensitivity, meaning and definition of environmental education, characteristics of environmental education, focal aspects of environmental education, goals and

objective of environmental education, need and importance of environmental education and scope of environmental education were discussed in detail.

## Review Questions

1. Define the term 'Environment'. (C) [Ans: 1:02]
2. Briefly explain the meaning of the term 'Environment' and enumerate its types. (B) [Ans: 1:01 + 1:03 + 1:03:1 + 1:03:2]
3. Briefly explain the types and components of environment. (A) [Ans: 1:01 + 1:03 + 1:03:1 + 1:03:1:01 + 1:03:1:02 + 1:03:2 + 1:03:3]
4. Explain the meaning of the term 'Environmental Awareness'. (B) [Ans: 1:04]
5. Explain the term 'Environmental Attitude' (C) [Ans: 1:05]
6. What do you understand by 'Ecological Intelligence'? (B) [Ans: 1:06]
7. What is meant by 'Ecological Sensitivity'? (C) [Ans: 1:07]
8. Define 'Environmental Education'. (C) [Ans: 1:05]
9. Explain the meaning and characteristics of Environmental Education (B) [Ans: 1:08 + 1:09]
10. Explain the focal aspects of Environmental Education (B) [Ans: 1:10]